

Specifications: Chess Tournament Management Application

1. Player Management

Register a Player

A member consists of:

- Username
- Email
- Date of Birth
- Gender (Male, Female, Other)
- ELO (Rating between 0 - 3000)

Business Rules:

- If a player does not have a rating (ELO), they start at **1200** by default.
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2. Tournament Management

Create a Tournament

A tournament consists of:

- Name and Location
- Number of Players (min: 2-32, max: 2-32)
- ELO (min: 0-3000, max: 0-3000)
- Category/ies (Junior, Senior, Veteran)
- Status (Waiting for players, In progress, Completed)
- Current Round Number
- Boolean **womenOnly** (Tournament reserved for Female/Other)

Business Rules:

- The minimum number of players must be $\leq$ the maximum number.
- The minimum ELO must be $\leq$ the maximum ELO.
- Upon creation: the current round is **0** and the status is **"Waiting for players"**.

Delete a Tournament

- **Business Rule:** A tournament can only be deleted if it has not yet started.
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3. Display and Consultation

Tournament List

Display the last 10 non-closed tournaments in descending order of update:

- Name, Location, Number of registrants, Min/Max players, Categories, Min/Max ELO, Status, Registration deadline, Current round.
- *Bonus*: Search filters (name, status, page, categories, etc.).

Tournament Details

- Present the same info as the list + the list of registered players.
 - *Bonus*: Display the matchups for the current round.
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4. Registrations and Cancellations

Add a Player to a Tournament

Business Rules (Conditions):

- The tournament has not yet started.
- The registration deadline has not passed.
- The player is not already registered.
- The maximum number of participants has not been reached.
- **Authorized Age**: * Junior: < 18 years old
 - Senior: ≥ 18 and < 60 years old
 - Veteran: ≥ 60 years old
- **Authorized ELO**: The score must be between the tournament's min and max ELO.
- **Gender**: If **womenOnly** is active, only "Female" and "Other" genders are admitted.

Unregister

- Possible if the tournament has not started and the player is currently registered.
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5. Tournament Progression

Start a Tournament

Business Rules:

- Minimum participants reached AND registration deadline passed.
- The current round moves to **1**.
- Matchup generation: **Round Robin (Double)**. Each player meets every other player twice (Home/Away).

Match Management

A match consists of: ID, Tournament ID, White Player, Black Player, Round Number, Result (Not played, White wins, Black wins, Draw).

- **Modification**: A match result can only be modified if it is part of the current round.

Move to Next Round

- Possible only if **all** matches of the current round have been played.
 - Increments the tournament's current round.
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6. Rankings and Scores

Display of the leaderboard (sorted by descending score):

- Player Name
- Number of matches played
- Number of wins, losses, and draws
- **Score Calculation:** Win = 1pt | Draw = 0.5pt | Loss = 0pt